Joseph E. Gonzalez, Associate Professor UC Berkeley Curriculum Vitae
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Biographic Summary

Joseph Gonzalez is an Associate Professor at UC Berkeley and co-director of the Berkeley Sky Computing Lab and the LMSYS Org, where he studies the design of next-generation cloud and AI systems for high-performance machine learning. Gonzalez's research spans AI systems, large language model (LLM) inference and serving, agentic AI, neural network design, and distributed systems for large-scale machine learning, with applications in natural language processing, computer vision, robotics, and autonomous driving. Gonzalez has published over 100 papers in top-tier venues spanning machine learning (NeurIPS, ICML, ICLR, AISTATS, UAI, CVPR, ECCV, ICCV) and computer systems (VLDB, OSDI, NSDI, EuroSys, MLSys, SoCC), which have been cited more than 80,000 times with an h-index of 102 (Google Scholar, 2026). He has made significant contributions to LLM serving systems (vLLM, SGLang), open LLM evaluation (LMSYS Chatbot Arena), the design of efficient neural networks for meta-learning, and prediction serving systems. Gonzalez co-developed and teaches the large upper-division data science class (Data100) and co-teaches Introduction to Machine Learning (CS189/289A) at UC Berkeley. Prior to joining Berkeley, Gonzalez co-founded Turi Inc. (formerly GraphLab) based on his thesis work and created the GraphX project (now part of Apache Spark). Gonzalez's innovative work has earned him significant recognition, including the Sigmod Systems Award, the NSF Early CAREER Award, and the Okawa Research Grant.

For more information visit `http://eecs.berkeley.edu/~jegonzal`.

Employment and Professional Experience

Nov 2020	— Present	Co-founder, RunLLM (formerly Aqueduct)
July 2021	— Present	Associate Professor in the EECS Department
Jan 2016	— July 2021	Assistant Professor in the EECS Department
May 2013	— July 2016	Co-founder at Turi Inc., Acquired by Apple Inc.
Jan 2013	— July 2015	Post-doc in the UC Berkeley AMPLab
June 2011	— Aug 2011	Yahoo! Research Intern
June 2007	— Aug 2007	AT&T Labs Intern

Education

Ph.D., Machine Learning: December 2012. Machine Learning Department, School of Computer Science at Carnegie Mellon University.

Title: *"Parallel and Distributed Algorithms and Systems for Probabilistic Reasoning."*

Thesis Advisor: Carlos Guestrin

M.S., Machine Learning: December 2009. Machine Learning Department, School of Computer Science at Carnegie Mellon University.

B.S. with Honors, Computer Science: June 2006. California Institute of Technology.

Awards

- **Diane S. McEntyre Award for Excellence in Teaching [2023]:** Award for excellence in teaching in the Department of Electrical Engineering and Computer Sciences at UC Berkeley.
- **Sigmod Systems Award [2022]:** Award for work on Apache Spark: an innovative, widely-used, open-source, unified data processing system encompassing relational, streaming, and machine-learning workloads.
- **NSF Early CAREER Award [2019]:** Funding awarded for research into the design of models and system for efficient prediction. Details: *"CAREER: Towards the Design of Models and Systems for Efficient Prediction"* (Award number 1846431).
- **Okawa Foundation Research Grant Award [2018]:** Funding award for research into the design of systems for machine learning life-cycle management.
- **Nominated for ACM Dissertation Award [2013]:** My thesis was nominated by CMU for the ACM Dissertation Award.
- **AT&T Labs Fellowship (2007):** Graduate research stipend for academic achievement as an under-represented minority.
- **NSF Graduate Research Fellowship (2007):** Graduate research stipend for 3 years.
- **NASA Space Act Award (2005):** Awarded for a sizeable contribution to space exploration.
- **NASA Inventions and Contributions Board Award (2005):** Awarded for the development of an innovative new technology that has made a contribution to space exploration.
- **Caltech Presidential Award (2002-2006):** I was awarded tuition for research and academic achievements.

Peer Reviewed Publications

[1] Parth Asawa, Alan Zhu, Abigail O'Neill, Matei Zaharia, Alexandros G. Dimakis, and Joseph E. Gonzalez. "How to Train Your Advisor: Steering Black-Box LLMs with Advisor Models". In: *International Conference on Machine Learning (ICML 2026)*. 2026. URL: https://arxiv.org/abs/2510.02453.

[2] Aditya Desai, Kunal Kumar Agrawal, Sehoon Yang, Alejandro Cuadron, Luis Gaspar Schroeder, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. "vAttention: Verified Sparse Attention via Sampling". In: *Proceedings of the Fourteenth International Conference on Learning Representations (ICLR 2026)*. 2026. URL: https://arxiv.org/abs/2510.05688.

[3] Shu Liu, Soujanya Ponnapalli, Shreya Shankar, Sepanta Zeighami, Alan Zhu, Shubham Agarwal, Ruiqi Chen, Samion Suwito, Shuo Yuan, Ion Stoica, Matei Zaharia, Alvin Cheung, Natacha Crooks, Joseph E. Gonzalez, and Aditya G. Parameswaran. "Supporting Our AI Overlords: Redesigning Data Systems to Be Agent-First". In: *Conference on Innovative Data Systems Research (CIDR)*. 2026. eprint: 2509.00997. URL: https://arxiv.org/abs/2509.00997.

[4] Michael Luo, Xiaoxiang Shi, Colin Cai, Tianjun Zhang, Justin Wong, Yichuan Wang, Chi Wang, Yanping Huang, Zhifeng Chen, Joseph E. Gonzalez, and Ion Stoica. "Autellix: An Efficient Serving Engine for LLM Agents as General Programs". In: *USENIX Symposium on Networked Systems Design and Implementation (NSDI)*. 2026. eprint: 2502.13965. URL: https://arxiv.org/abs/2502.13965.

[5] Qiuyang Mang, Wenhao Chai, Zhifei Li, Huanzhi Mao, Shang Zhou, Alexander Du, Hanchen Li, Shu Liu, Edwin Chen, Yichuan Wang, Xieting Chu, Zerui Cheng, Yuan Xu, Tian Xia, Zirui Wang, Tianneng Shi, Jianzhu Yao, Yilong Zhao, Qizheng Zhang, Charlie Ruan, Zeyu Shen, Kaiyuan Liu, Zhaoyang Hong, Alex Gu, Ziyi Zhang, Runyuan He, Dong Xing, Zerui Li, Zirong Zeng, Yige Jiang, Lufeng Cheng, Ziyi Zhao, Youran Sun, Suyang Zhong, Junpeng Wang, Donglin Li, Wenyuan Huang, Jialiang Gu, Wesley Zheng, Wangmeiyu Zhang, Ruyi Ji, Xuechang Tu, Zihan Zheng, Zhaozi Wang, Zexing Chen, Jingbang Chen, Jialu Zhang, Aleksandra Korolova, Peter Henderson, Pramod Viswanath, Vijay Ganesh, Saining Xie, Zhuang Liu, Dawn Song, Sewon Min, Ion Stoica, Joseph E. Gonzalez, Jingbo Shang, and Alvin Cheung. "FrontierCS: Evolving Challenges for Evolving Intelligence". In: *International Conference on Machine Learning (ICML 2026)*. 2026. URL: https://arxiv.org/abs/2512.15699.

[6] Huanzhi Mao, Aditya Ghai, Imra Dawoodani, Tony Ginart, Shishir G. Patil, John Emmons, and Joseph E. Gonzalez. "MFCL Audio: An Audio Function Calling Evaluation for Large Language Models". In: *International Conference on Machine Learning (ICML 2026)*. 2026. URL: https://icml.cc/virtual/2026/poster/61489.

[7] Mihran Miroyan, Tsung-Han Wu, Logan King, Tianle Li, Jiayi Pan, Xinyan Hu, Wei-Lin Chiang, Anastasios N. Angelopoulos, Trevor Darrell, Narges Norouzi, and Joseph E. Gonzalez. "Search Arena: Analyzing Search-Augmented LLMs". In: *International Conference on Learning Representations (ICLR 2026)*. 2026. URL: https://arxiv.org/abs/2506.05334.

[8] Melissa Z. Pan, Negar Arabzadeh, Riccardo Cogo, Yuxuan Zhu, Alexander Xiong, Lakshya A. Agrawal, Huanzhi Mao, Emma Shen, Sid Pallerla, Liana Patel, Shu Liu, Tianneng Shi, Xiaoyuan Liu, Jared Quincy Davis, Emmanuele Lacavalla, Alessandro Basile, Shuyi Yang, Paul Castro, Daniel Kang, Koushik Sen, Dawn Song, Joseph E. Gonzalez, Ion Stoica, Matei Zaharia, and Marquita Ellis. "Characterizing Agents in Production". In: *International Conference on Machine Learning (ICML 2026)*. 2026. URL: https://arxiv.org/abs/2512.04123.

[9] Shishir G. Patil, Saurav Kumar, Prabal Dutta, and Joseph E. Gonzalez. "Efficient ML Model Updates for Deeply Embedded Microcontrollers". In: *Proceedings of the 21st European Conference on Computer Systems (EuroSys 2026)*. 2026.

[10] Yifan Qiao, Shan Yu, Shu Anzai, Haoran Ma, Shuo Yang, Yang Wang, Miryung Kim, Yongji Wu, Yang Zhou, Jiarong Xing, Joseph E. Gonzalez, Ion Stoica, and Harry Xu. "ConServe: Fine-Grained GPU Harvesting for LLM Online and Offline Co-Serving". In: *International Conference on Machine Learning (ICML 2026)*. 2026. URL: https://arxiv.org/abs/2410.01228.

[11] Luis Gaspar Schroeder, Aditya Desai, Alejandro Cuadron, Kyle Chu, Shu Liu, Mark Zhao, Stephan Krusche, Alfons Kemper, Matei Zaharia, and Joseph E. Gonzalez. "vCache: Verified Semantic Prompt Caching". In: *International Conference on Learning Representations (ICLR 2026)*. 2026. URL: https://arxiv.org/abs/2502.03771.

[12] Tsung-Han Wu, Mihran Miroyan, David M. Chan, Trevor Darrell, Narges Norouzi, and Joseph E. Gonzalez. "Are Large Reasoning Models Interruptible?" In: *International Conference on Machine Learning (ICML 2026)*. 2026. URL: https://arxiv.org/abs/2510.11713.

[13] Jintao Zhang, Haoxu Wang, Kai Jiang, Shuo Yang, Kaiwen Zheng, Haocheng Xi, Ziteng Wang, Hongzhou Zhu, Min Zhao, Ion Stoica, Joseph E. Gonzalez, Jianfei Chen, and Jun Zhu. "SLA: Beyond Sparsity in Diffusion Transformers via Fine-Tunable Sparse-Linear Attention". In: *International Conference on Learning Representations (ICLR 2026)*. 2026. URL: https://arxiv.org/abs/2509.24006.

[14] Asim Biswal, Liana Patel, Siddarth Jha, Amog Kamsetty, Shu Liu, Joseph E. Gonzalez, Carlos Guestrin, and Matei Zaharia. "Text2SQL Is Not Enough: Unifying AI and Databases with TAG". In: *Conference on Innovative Data Systems Research (CIDR)*. 2025. eprint: 2408.14717. URL: https://arxiv.org/abs/2408.14717.

[15] Shiyi Cao, Shu Liu, Tyler Griggs, Peter Schafhalter, Xiaoxuan Liu, Ying Sheng, Joseph E. Gonzalez, Matei Zaharia, and Ion Stoica. "MoE-Lightning: High-Throughput MoE Inference on Memory-Constrained GPUs". In: *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2025)*. 2025. URL: https://arxiv.org/abs/2411.11217.

[16] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. "Why Do Multi-Agent LLM Systems Fail?" In: *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. 2025. URL: https://arxiv.org/abs/2503.13657.

[17] Christopher Chou, Lisa Dunlap, Koki Mashita, Krishna Mandal, Trevor Darrell, Ion Stoica, Joseph E. Gonzalez, and Wei-Lin Chiang. "VisionArena: 230K Real-World User-VLM Conversations with Preference Labels". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025)*. 2025. URL: https://arxiv.org/abs/2412.08687.

[18] Aditya Desai, Shuo Yang, Alejandro Cuadron, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. "HashAttention: Semantic Sparsity for Faster Inference". In: *International Conference on Machine Learning (ICML)*. 2025. eprint: 2412.14468. URL: https://arxiv.org/abs/2412.14468.

[19] Lisa Dunlap, Joseph E. Gonzalez, Trevor Darrell, Fabian Caba Heilbron, Josef Sivic, and Bryan Russell. "Discovering Divergent Representations between Text-to-Image Models". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV 2025)*. 2025. URL: https://arxiv.org/abs/2509.08940.

[20] Lisa Dunlap, Krishna Mandal, Trevor Darrell, Jacob Steinhardt, and Joseph E. Gonzalez. "VibeCheck: Discover and Quantify Qualitative Differences in Large Language Models". In: *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR 2025)*. 2025. URL: https://arxiv.org/abs/2410.12851.

[21] Evan Frick, Tianle Li, Connor Chen, Wei-Lin Chiang, Anastasios Angelopoulos, Jiantao Jiao, Banghua Zhu, Joseph E. Gonzalez, and Ion Stoica. "How to Evaluate Reward Models for RLHF". In: *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR 2025)*. 2025. URL: https://arxiv.org/abs/2410.14872.

[22] Dacheng Li, Shiyi Cao, Chengkun Cao, Xiuyu Li, Shangyin Tan, Kurt Keutzer, Jiarong Xing, Joseph E. Gonzalez, and Ion Stoica. "S*: Test Time Scaling for Code Generation". In: *Findings of the Association for Computational Linguistics: EMNLP 2025*. 2025. eprint: 2502.14382. URL: https://arxiv.org/abs/2502.14382.

[23] Dacheng Li, Shiyi Cao, Tyler Griggs, Shu Liu, Xiangxi Mo, Eric Tang, Sumanth Hegde, Kourosh Hakhamaneshi, Shishir G. Patil, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. "Language Models Can Easily Learn to Reason from Demonstrations: Structure, Not Content, Is What Matters!" In: *Findings of the Association for Computational Linguistics: EMNLP 2025*. 2025. eprint: 2502.07374. URL: https://arxiv.org/abs/2502.07374.

[24] Dacheng Li, Yunhao Fang, Yukang Chen, Shuo Yang, Shiyi Cao, Justin Wong, Michael Luo, Xiaolong Wang, Hongxu Yin, Joseph E. Gonzalez, Ion Stoica, Song Han, and Yao Lu. "WorldModelBench: Judging Video Generation Models as World Models". In: *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. 2025. URL: https://arxiv.org/abs/2502.20694.

[25] Tianle Li, Wei-Lin Chiang, Evan Frick, Lisa Dunlap, Tianhao Wu, Banghua Zhu, Joseph E. Gonzalez, and Ion Stoica. "From Crowdsourced Data to High-Quality Benchmarks: Arena-Hard and BenchBuilder Pipeline". In: *International Conference on Machine Learning (ICML)*. 2025. eprint: 2406.11939. URL: https://arxiv.org/abs/2406.11939.

[26] Shu Liu, Asim Biswal, Amog Kamsetty, Audrey Cheng, Luis Gaspar Schroeder, Liana Patel, Shiyi Cao, Xiangxi Mo, Ion Stoica, Joseph E. Gonzalez, and Matei Zaharia. "Optimizing LLM Queries in

Relational Data Analytics Workloads". In: *Proceedings of Machine Learning and Systems 7 (MLSys 2025)*. 2025. URL: https://arxiv.org/abs/2403.05821.

[27] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. "RouteLLM: Learning to Route LLMs with Preference Data". In: *International Conference on Learning Representations (ICLR)*. 2025. eprint: 2406.18665. URL: https://arxiv.org/abs/2406.18665.

[28] Shishir G. Patil, Hanchen Mao, Fanyue Yan, Chung-Cheng Jack Ji, Vivek Suresh, Ion Stoica, and Joseph E. Gonzalez. "The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models". In: *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*. 2025. URL: https://gorilla.cs.berkeley.edu/blogs/8_berkeley_function_calling_leaderboard.html.

[29] Justin Wong, Yury Orlovskiy, Michael Luo, Sanjit A. Seshia, and Joseph E. Gonzalez. "SimpleStrat: Diversifying Language Model Generation with Stratification". In: *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. 2025. URL: https://arxiv.org/abs/2410.09038.

[30] Tsung-Han Wu, Giscard Biamby, Jerome Quenum, Ritwik Gupta, Joseph E. Gonzalez, Trevor Darrell, and David M. Chan. "Visual Haystacks: A Vision-Centric Needle-in-a-Haystack Benchmark". In: *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR 2025)*. 2025. URL: https://arxiv.org/abs/2407.13766.

[31] Tsung-Han Wu, Joseph E. Gonzalez, Trevor Darrell, and David M. Chan. "CLAIRA: Leveraging Large Language Models to Judge Audio Captions". In: *2025 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU 2025)*. 2025. URL: https://arxiv.org/abs/2409.12962.

[32] Tsung-Han Wu, Heekyung Lee, Jiaxin Ge, Joseph E. Gonzalez, Trevor Darrell, and David M. Chan. "Generate, but Verify: Reducing Hallucination in Vision-Language Models with Retrospective Resampling". In: *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. 2025. URL: https://arxiv.org/abs/2504.13169.

[33] Ling Yang, Zhaochen Yu, Tianjun Zhang, Minkai Xu, Joseph E. Gonzalez, Bin Cui, and Shuicheng Yan. "SuperCorrect: Advancing Small LLM Reasoning with Thought Template Distillation and Self-Correction". In: *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR 2025)*. 2025. URL: https://arxiv.org/abs/2410.09008.

[34] Chen Zhang, Kuntai Du, Shu Liu, Woosuk Kwon, Xiangxi Mo, Yufeng Wang, Xiaoxuan Liu, Kaichao You, Zhuohan Li, Mingsheng Long, Jidong Zhai, Joseph E. Gonzalez, and Ion Stoica. "JENGA: Effective Memory Management for Serving LLM with Heterogeneity". In: *Proceedings of the ACM SIGOPS 31st Symposium on Operating Systems Principles (SOSP 2025)*. 2025. URL: https://arxiv.org/abs/2503.18292.

[35] Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E. Gonzalez, and Ion Stoica. "Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference". In: *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*. 2024. URL: https://arxiv.org/abs/2403.04132.

[36] Lisa Dunlap, Yuhui Zhang, Xiaohan Wang, Ruiqi Zhong, Trevor Darrell, Jacob Steinhardt, Joseph E. Gonzalez, and Serena Yeung-Levy. "Describing Differences in Image Sets with Natural Language". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*. 2024. URL: https://arxiv.org/abs/2312.02974.

[37] Naman Jain, Tianjun Zhang, Wei-Lin Chiang, Joseph E. Gonzalez, Koushik Sen, and Ion Stoica. "LLM-Assisted Code Cleaning for Training Accurate Code Generators". In: *Proceedings of the Twelfth International Conference on Learning Representations (ICLR 2024)*. 2024. URL: https://arxiv.org/abs/2311.14904.

[38] Dacheng Li, Rulin Shao, Anze Xie, Eric P. Xing, Xuezhe Ma, Ion Stoica, Joseph E. Gonzalez, and Hao Zhang. "DistFlashAttn: Distributed Memory-Efficient Attention for Long-Context LLMs Training". In: *Conference on Language Modeling (COLM)*. 2024. eprint: 2310.03294. URL: https://arxiv.org/abs/2310.03294.

[39] Michael Luo, Justin Wong, Brandon Trabucco, Yanping Huang, Joseph E. Gonzalez, Zhifeng Chen, Ruslan Salakhutdinov, and Ion Stoica. "Stylus: Automatic Adapter Selection for Diffusion Models". In: *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. 2024. URL: https://arxiv.org/abs/2404.18928.

[40] Federico Mora, Justin Wong, Haley Lepe, Sahil Bhatia, Karim Elmaaroufi, George Varghese, Joseph E. Gonzalez, Elizabeth Polgreen, and Sanjit A. Seshia. "Synthetic Programming Elicitation for Text-to-Code in Very Low-Resource Programming and Formal Languages". In: *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. 2024. URL: https://arxiv.org/abs/2406.03636.

[41] Shishir G. Patil, Tianjun Zhang, Xin Wang, and Joseph E. Gonzalez. "Gorilla: Large Language Model Connected with Massive APIs". In: *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. 2024. eprint: 2305.15334. URL: https://arxiv.org/abs/2305.15334.

[42] Suzanne Petryk, David M. Chan, Anish Kachinthaya, Haodi Zou, John Canny, Joseph E. Gonzalez, and Trevor Darrell. "ALOHA: A New Measure for Hallucination in Captioning Models". In: *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024)*. 2024. URL: https://arxiv.org/abs/2404.02904.

[43] Suzanne Petryk, Spencer Whitehead, Joseph E. Gonzalez, Trevor Darrell, Anna Rohrbach, and Marcus Rohrbach. "Simple Token-Level Confidence Improves Caption Correctness". In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2024. eprint: 2305.07021. URL: https://arxiv.org/abs/2305.07021.

[44] Daniel Rothchild, Andrew S. Rosen, Eric Taw, Connie Robinson, Joseph E. Gonzalez, and Aditi S. Krishnapriyan. "Investigating the Behavior of Diffusion Models for Accelerating Electronic Structure Calculations". In: *Chemical Science* (2024). URL: https://arxiv.org/abs/2311.01491.

[45] Sheng Shen, Shijia Yang, Tianjun Zhang, Bohan Zhai, Joseph E. Gonzalez, Kurt Keutzer, and Trevor Darrell. "Multitask Vision-Language Prompt Tuning". In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2024)*. 2024. URL: https://arxiv.org/abs/2211.11720.

[46] Ying Sheng, Shiyi Cao, Dacheng Li, Coleman Hooper, Nicholas Lee, Shuo Yang, Christopher Chou, Banghua Zhu, Lianmin Zheng, Kurt Keutzer, Joseph E. Gonzalez, and Ion Stoica. "SLoRA: Scalable Serving of Thousands of LoRA Adapters". In: *Proceedings of Machine Learning and Systems 6 (MLSys 2024)*. 2024. URL: https://arxiv.org/abs/2311.03285.

[47] Ying Sheng, Shiyi Cao, Dacheng Li, Banghua Zhu, Zhuohan Li, Danyang Zhuo, Joseph E. Gonzalez, and Ion Stoica. "Fairness in Serving Large Language Models". In: *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 2024)*. 2024. URL: https://arxiv.org/abs/2401.00588.

[48] Sijun Tan, Xiuyu Li, Shishir G. Patil, Ziyang Wu, Tianjun Zhang, Kurt Keutzer, Joseph E. Gonzalez, and Raluca Ada Popa. "LLoCO: Learning Long Contexts Offline". In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP 2024)*. 2024. URL: https://arxiv.org/abs/2404.07979.

[49] Sarah Wooders, Shishir Liu, Paras Jain, Xiangxi Mo, Joseph E. Gonzalez, Vincent Liu, and Ion Stoica. "CloudCast: High-Throughput, Cost-Aware Overlay Multicast in the Cloud". In: *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI 2024)*. 2024. URL: https://www.usenix.org/conference/nsdi24/presentation/wooders.

[50] Tsung-Han Wu, Giscard Biamby, David Chan, Lisa Dunlap, Ritwik Gupta, Xudong Wang, Joseph E. Gonzalez, and Trevor Darrell. "See, Say, and Segment: Teaching LMMs to Overcome False Premises". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*. 2024. URL: https://arxiv.org/abs/2312.08366.

[51] Tsung-Han Wu, Long Lian, Joseph E. Gonzalez, Boyi Li, and Trevor Darrell. "Self-Correcting LLM-Controlled Diffusion Models". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)*. 2024. URL: https://arxiv.org/abs/2311.16090.

[52] Jiezhi Yang, Khushi Desai, Charles Packer, Harshil Bhatia, Nicholas Rhinehart, Rowan McAllister, and Joseph E. Gonzalez. "CARFF: Conditional Auto-Encoded Radiance Field for 3D Scene Forecasting". In: *Proceedings of the European Conference on Computer Vision (ECCV 2024)*. 2024. URL: https://arxiv.org/abs/2401.18075.

[53] Ling Yang, Zhaochen Yu, Tianjun Zhang, Shiyi Cao, Minkai Xu, Wentao Zhang, Joseph E. Gonzalez, and Bin Cui. "Buffer of Thoughts: Thought-Augmented Reasoning with Large Language Models". In: *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. 2024. URL: https://arxiv.org/abs/2406.04271.

[54] Tianjun Zhang, Shishir G. Patil, Naman Jain, Sheng Shen, Matei Zaharia, Ion Stoica, and Joseph E. Gonzalez. "RAFT: Adapting Language Model to Domain Specific RAG". In: *Conference on Language Modeling (COLM)*. 2024. eprint: 2403.10131. URL: https://arxiv.org/abs/2403.10131.

[55] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Tianle Li, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zhuohan Li, Zi Lin, Eric P. Xing, Joseph E. Gonzalez, Ion Stoica, and Hao Zhang. "LMSYS-Chat-1M: A Large-Scale Real-World LLM Conversation Dataset". In: *Proceedings of the Twelfth International Conference on Learning Representations (ICLR 2024)*. 2024. URL: https://arxiv.org/abs/2309.11998.

[56] Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. "SGLang: Efficient Execution of Structured Language Model Programs". In: *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. 2024. URL: https://arxiv.org/abs/2312.07104.

[57] Robert Avram, Jeffrey E. Olgin, Zeeshan Ahmed, Louis Verreault-Julien, Alvin Wan, Julia Barrios, Justin Wan, Joseph E. Gonzalez, Derek So, Krishan Soni, and Geoffrey H. Tison. "CathAI: Fully Automated Coronary Angiography Interpretation and Stenosis Estimation". In: *npj Digital Medicine*. 2023. URL: https://www.nature.com/articles/s41746-023-00880-1.

[58] David M. Chan, Suzanne Petryk, Joseph E. Gonzalez, Trevor Darrell, and John Canny. "CLAIR: Evaluating Image Captions with Large Language Models". In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*. 2023. URL: https://arxiv.org/abs/2310.12971.

[59] Lisa Dunlap, Alyssa Umino, Han Zhang, Jiezhi Yang, Joseph E. Gonzalez, and Trevor Darrell. "Diversify Your Vision Datasets with Automatic Diffusion-Based Augmentation". In: *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. 2023. eprint: 2305.16289. URL: https://arxiv.org/abs/2305.16289.

[60] Joseph E. Gonzalez and Yucheng Low. "The Story of GraphLab—From Scaling Machine Learning to Shaping Graph Systems Research (VLDB 2023 Test-of-Time Award Talk)". In: *Proceedings of the VLDB Endowment*. 2023. URL: https://dl.acm.org/doi/10.14778/3611479.3611538.

[61] Kirthevasan Kandasamy, Joseph E. Gonzalez, Michael I. Jordan, and Ion Stoica. "VCG Mechanism Design with Unknown Agent Values under Stochastic Bandit Feedback". In: *Journal of Machine Learning Research* 24 (2023). URL: https://arxiv.org/abs/2004.08924.

[62] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. "Efficient Memory Management for Large Language Model

Serving with PagedAttention". In: *Proceedings of the 29th ACM Symposium on Operating Systems Principles (SOSP 2023)*. 2023. URL: https://arxiv.org/abs/2309.06180.

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[9] Alexander Krentsel, Peter Schafhalter, Joseph E. Gonzalez, Sylvia Ratnasamy, Scott Shenker, and Ion Stoica. *Managing Bandwidth: The Key to Cloud-Assisted Autonomous Driving*. 2024. eprint: 2410.16227. URL: https://arxiv.org/abs/2410.16227.

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[20] Brijen Thananjeyan, Kirthevasan Kandasamy, Ion Stoica, Michael I. Jordan, Ken Goldberg, and Joseph E. Gonzalez. *PAC Best Arm Identification Under a Deadline*. 2021. DOI: 10.48550/ARXIV.2106.03221. URL: https://arxiv.org/abs/2106.03221.

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[34] Joseph E. Gonzalez, Peter Bailis, Michael I. Jordan, Michael J. Franklin, Joseph M. Hellerstein, Ali Ghodsi, and Ion Stoica. "Asynchronous Complex Analytics in a Distributed Dataflow Architecture". In: *CoRR* abs/1510.07092 (2015). arXiv: 1510.07092. URL: http://arxiv.org/abs/1510.07092.

Other Technical Reports and Invited Publications

[1] David Patterson, Joseph Gonzalez, Urs Hölzle, Quoc Le, Chen Liang, Lluis-Miquel Munguia, Daniel Rothchild, David R. So, Maud Texier, and Jeff Dean. "The Carbon Footprint of Machine Learning Training Will Plateau, Then Shrink". In: *Computer* 55.7 (2022), pp. 18–28. DOI: 10.1109/MC.2022.3148714. URL: https://arxiv.org/abs/2204.05149.

[2] David E. Culler, Prabal Dutta, Gabe Fierro, Joseph E. Gonzalez, Nathan Pemberton, Johann Schleier-Smith, Kalyanaraman Shankari, Alvin Wan, and Thomas Zachariah. "CoVista: A Unified View on Privacy Sensitive Mobile Contact Tracing". In: *IEEE Data Eng. Bull.* 43.2 (2020), pp. 83–94. URL: http://sites.computer.org/debull/A20june/p83.pdf.

[3] Eric Jonas, Johann Schleier-Smith, Vikram Sreekanti, Chia-Che Tsai, Anurag Khandelwal, Qifan Pu, Vaishaal Shankar, Joao Menezes Carreira, Karl Krauth, Neeraja Yadwadkar, Joseph E. Gonzalez, Raluca Ada Popa, Ion Stoica, and David A. Patterson. *Cloud Programming Simplified: A Berkeley View on Serverless Computing*. Tech. rep. UCB/EECS-2019-3. EECS Department, University of California, Berkeley, Feb. 2019. URL: http://www2.eecs.berkeley.edu/Pubs/TechRpts/2019/EECS-2019-3.html.

[4] Dan Crankshaw, Joseph E. Gonzalez, and Peter Bailis. "Research for Practice: Prediction-Serving Systems". In: *Commun. ACM* 61.8 (July 2018), pp. 45–49. ISSN: 0001-0782. URL: http://doi.acm.org/10.1145/3190574.

[5] Ion Stoica, Dawn Song, Raluca Ada Popa, David A. Patterson, Michael W. Mahoney, Randy H. Katz, Anthony D. Joseph, Michael Jordan, Joseph M. Hellerstein, Joseph E. Gonzalez, Ken Goldberg, Ali Ghodsi, David E. Culler, and Pieter Abbeel. *A Berkeley View of Systems Challenges for AI*. Tech. rep. UCB/EECS-2017-159. EECS Department, University of California, Berkeley, Sept. 2017. URL: http://www2.eecs.berkeley.edu/Pubs/TechRpts/2017/EECS-2017-159.html.

[6] Rong Gu, Qianhao Dong, Haoyuan Li, Joseph E. Gonzalez, Zhao Zhang, Shuai Wang, Yihua Huang, Scott Shenker, Ion Stoica, and Patrick P. C. Lee. *DFS-Perf: A Scalable and Unified Benchmarking Framework for Distributed File Systems*. Tech. rep. UCB/EECS-2016-133. EECS Department, University of California, Berkeley, July 2016. URL: http://www2.eecs.berkeley.edu/Pubs/TechRpts/2016/EECS-2016-133.html.

[7] Matei Zaharia, Reynold S. Xin, Patrick Wendell, Tathagata Das, Michael Armbrust, Ankur Dave, Xiangrui Meng, Josh Rosen, Shivaram Venkataraman, Michael J. Franklin, Ali Ghodsi, Joseph E. Gonzalez, Scott Shenker, and Ion Stoica. "Apache Spark: A Unified Engine for Big Data Processing". In: *Commun. ACM* 59.11 (Sept. 2016), pp. 56–65. ISSN: 0001-0782. URL: https://doi.acm.org/10.1145/2934664.

[8] Joseph E. Gonzalez. "From Graphs to Tables the Design of Scalable Systems for Graph Analytics". In: *Proceedings of the 23rd International Conference on World Wide Web*. WWW '14 Companion. Seoul, Korea: ACM, 2014, pp. 1149–1150. ISBN: 978-1-4503-2745-9. URL: https://doi.acm.org/10.1145/2567948.2580059.

Journal Editor

- **[2020] Editor of the IEEE Data Engineering Bulletin Special Issue:** Data Technologies Behind Digital Contact Tracing
- **[2018] Editor of the IEEE Data Engineering Bulletin Special Issue:** Machine Learning Lifecycle Management

Selected Invited Talks

- **[2026] Invited Keynote at the Data Discovery Symposium** (UC Berkeley, May 2026) *Invited keynote at this interdisciplinary symposium on data-driven discovery.*
- **[2026] Invited Talk at Samsung Research America AI Summit** (San Jose, May 2026) *Finding a New North Star for AI Research in the Post AGI Era*: Examining how scaling systems and humans enables capable AI agents.
- **[2026] Invited Faculty Talk at the Amazon AI PhD Fellowship Summit** (Seattle, April 2026) *An Algorithm for Innovation*: Invited faculty speaker at the annual Amazon AI Fellowship gathering of PhD fellows and faculty advisors.
- **[2025] Invited Keynote at the Samsung AI Forum** (Online, September 2025) *Invited keynote talk on advances in AI systems research for Samsung's annual AI research forum.*
- **[2022] Invited Speaker at the Cloud Intelligence and AIOps Workshop** *The Evolution of Model Deployment from the Cloud to the Sky*
- **[2022] Invited Speaker at the China Database Committee Conference** *The Co-Evolution of Data and ML Systems from the Cloud to the Sky*
- **[2021] UC Berkeley Office of the CTO Cloud Computing Meetup** *To the Cloud and Beyond A Brief View of Cloud Computing Research at UC Berkeley*
- **[2020] Speaker in the Colloquium Lecture Series hosted by the Department of Computer and Information Science at the University of Pennsylvania** *Advances in Machine Learning Systems Research*
- **[2020] Keynote Speaker at the 14th Annual Machine Learning Symposium at the New York Academy of Sciences** *Advances in AI-Systems Research*
- **[2020] Interviewed on This Week in Machine Learning Podcast**
- **[2020] Keynote at the International Solid-State Circuits Conference Forum on Machine Learning Processors** *Hardware Opportunities in the Machine Learning Lifecycle*

- **[2020] Keynote at the Workshop on MLOps (co-located and MLSys'20)**: *Advances in Prediction Serving Systems*
- **[2019] Spoke in a Panel for Hispanic Engineers and Scientists (HES)**: *Discussed career planning and my research trajectory to undergrads at UC Berkeley.*
- **[2018] Data Eng Conference Keynote** *Managing the Machine Learning Lifecycle*
- **[2018] ISAT TAMALE Workshop Presentation:** *This invitation only workshop covered the Tools and Algorithms for Machine learning. Gonzalez presented his work on prediction serving and experiment management systems.*
- **[2017] SOSP Workshop Keynote** *Research at the Intersection of AI and Systems*
- **[2017] AAAI Systems Workshop Keynote** *Rise to the Challenges of AI Systems*
- **[2016] ICML Workshop Keynote** *Prediction Serving: What happens after learning?*
- **[2015] NIPS Workshop Keynote** *Intelligent Services: Serving Machine Learning Predictions*
- **[2015] ODSC** *Intelligent Services: Serving Machine Learning Predictions*
- **[2015] CIDR Talk** *The Missing Piece in Complex Analytics: Low Latency, Scalable Model Management and Serving with Velox*
- **[2014] OSDI Conference** *GraphX: Graph Processing in a Distributed Dataflow Framework*
- **[2014] Annual meeting of the International Society for Bayesian Analysis (ISBA)** *Concurrency Control For Scalable Bayesian Inference.*
- **[2014] Tutorial at the International Conference for Machine Learning (ICML)** *Emerging Systems for Large-Scale Machine Learning.*
- **[2014] Session on Graph Algorithms Building Blocks at the International Parallel and Distributed Processing Systems (IPDPS)** *GraphX: Unifying Table and Graph Analytics.*
- **[2014] NetApp ATG University Day** *Large Scale Graph Analytics: Applications and Systems*
- **[2014] Keynote Speaker: Workshop on Big Graph Mining at the International World Wide Web Conference (WWW)** *From Graphs to Tables: The Design of Scalable Systems for Graph Analytics.*
- **[2013] SIAM CSE'13 Minisymposium Frontiers in Large-Scale Graph Analysis** Large-Scale Graph-Structured Machine Learning: GraphLab in the Cloud and GraphChi in your PC
- **[2012] OSDI Conference** *PowerGraph: Distributed Graph-Parallel Computation on Natural Graphs*
- **[2011] IDGA Data Center Consolidation Summit** *GraphLab: A New Parallel Framework for Machine Learning*
- **[2010] DARPA Future Ideas Symposium** Invited speaker at the DARPA future ideas symposium.
- **[2009] UAI Conference.** *Distributed Parallel Inference on Large Factor Graphs*

Advising

Postdocs

1. Jianfei Chen [2019-2021] *Tsinghua University*
2. Aditya Desai [2024-2026] *IIT Bombay (Assistant Professor)*
3. Ionel Gog [2018-2021] *Google Brain*
4. Kirthevasan Kandasamy [2019-2021] *UW-Madison*
5. Yifan Qiao [2024-2025] *Inferact (Staff Engineer)*
6. Alexey Tumanov [2016-2019] *Georgia Tech.*
7. Matthew Wright [2019-2021]
8. Yaoqing Yang [2019-2022] *Dartmouth*

PhD Candidates

1. Francois Belletti (PhD) [2015-2018]: **Thesis:** *Alternate Representations for Scalable Analysis and Control of Heterogeneous Time Series*
2. Asim Biswal [2023-Present]: *Studying LLM-powered query processing and optimization for relational data analytics.*
3. Shiyi Cao [2023-Present]: *Studying efficient LLM inference and serving systems, including speculative decoding and mixture-of-experts optimization.*
4. Daniel Crankshaw (PhD) [2015-2019]: **Thesis:** *The Design and Implementation of Low-Latency Prediction Serving Systems Google (Staff Software Engineer).*
5. Lisa Dunlap (PhD) [2021-2026]: **Thesis:** *So What's the Vibe? Data-Driven Diagnosis of Model Behavior Anthropic.*
6. Vlad Feinberg [2017-2018]: *Departed without degree; Distinguished Engineer at Google DeepMind and Gemini pre-training lead.*
7. Guangyu Feng [2025-Present]: *Studying asynchronous patterns for agentic tool use.*
8. Yu Gai (PhD) [2019-]: *Studied graph neural networks for question answering.*
9. Paras Jain (PhD) [2018-2023]: **Thesis:** *Towards Robust and Scalable Large Language Models Genmo (CEO & co-founder).*
10. Dacheng Li [2023-Present]: *Studying AI systems for large-scale LLM serving and evaluation, with contributions to the LMSYS ecosystem and world model benchmarking.*
11. Hanchen Li [2025-Present]: *Studying agent-first data and serving systems, including KV-cache-aware scheduling for multi-turn LLM agents.*
12. Richard Liaw[2017-2020]: *Studying the design of systems for hyperparameter search.*
13. Charles Lin [2019-2020]: *Studying economics of serverless computing.*
14. Kevin Lin (PhD) [2019-2024]: **Thesis:** *Structured Contexts For Large Language Models Letta.*
15. Charles Packer (PhD) [2019-2024]: **Thesis:** *Building Agentic Systems in an Era of Large Language Models Letta (CEO & co-founder).*
16. Shishir Patil (PhD) [2019-2024]: **Thesis:** *Teaching Large Language Models to Use Tools at Scale Meta.*
17. Nathan Pemberton [2020-2022]: **Thesis:** *The Serverless Datacenter: Hardware and Software Techniques for Resource Disaggregation Amazon (Applied Scientist).*

18. Devin Petersohn [2015-2021]: **Thesis:** *Dataframe Systems: Theory, Architecture, and Implementation Snowflake.*

19. Suzanne Petryk (PhD) [2019-2024]: **Thesis:** *Reliable Multimodal Models Apple.*

20. Daniel Rothchild (PhD) [2018-2023]: **Thesis:** *Accelerating Electronic Structure Calculations with Machine Learning Genentech.*

21. Peter Schafhalter (PhD) [2019-2025]: **Thesis:** *Towards Cloud-Assisted Autonomous Driving Microsoft AI (Technical Staff).*

22. Minhyuk Seo [2026-Present]: *Studying continual and collaborative learning for human-like AI.*

23. Eyal Sela (PhD) [2018-2026]: **Thesis:** *Algorithmic Methods For Efficient Deep Learning Inference on the Edge*

24. Vikram Sreekanti [2016-2020]: **Thesis:** *The Design of Stateful Serverless Infrastructure Herald.ai (CEO).*

25. Brijen Thananjeyan [2018-2022]: **Thesis:** *Safe Reinforcement Learning Using Learned Safe Sets Google DeepMind (Senior Research Scientist).*

26. Alvin Wan (PhD) [2018-2022]: **Thesis:** *Efficiently Designing Efficient Deep Neural Networks OpenAI.*

27. Xin Wang (PhD) [2016-2020]: **Thesis:** *The Design of Dynamic Neural Networks for Efficient Learning and Inference OpenAI (Technical Staff).*

28. Colin Wang [2025-Present]: *Studying multimodal AI systems and polylithic agent architectures.*

29. Justin Wong (PhD) [2020-2026]: **Thesis:** *Mitigating Post-Training Effects on Generative Diversity in Language Models Amazon.*

30. Yichuan Wang [2024-Present]: *Studying storage-efficient vector indexing and RAG systems for personal devices. Learning based approaches to accelerate combinatorial optimization problems.*

31. Sarah Wooders (PhD) [2020-2024]: **Thesis:** *Ensuring Data Freshness Across Clouds for Model Serving Letta (CTO & co-founder).*

32. Tsung-Han (Patrick) Wu [2023-Present]: *Studying multimodal models and their evaluation, including hallucination reduction, false-premise correction, and vision-centric benchmarks.*

33. Parth Asawa [2024-Present]: *Studying continual learning and sample-efficient model adaptation.*

34. Huanzhi Mao [2025-Present]: *Studying LLM tool use and function calling; leads the Berkeley Function Calling Leaderboard (BFCL).*

35. Mihran Miroyan [2024-Present]: *Studying evaluation and benchmarking of search-augmented large language models.*

36. Mayank Mishra [2025-Present]: *Studying efficient LLM pretraining and novel model architectures for scalable language modeling.*

37. Xiangxi (Simon) Mo (PhD) [2023-2026]: **Thesis:** *Building Open Source Inference Serving Systems Inferact (CEO & co-founder).*

38. Abby O'Neill [2025-Present]: *Studying AI–AI interactions and cooperative multi-agent system design.*

39. Neeraja Yadwadkar (PhD) [2015-2018]: **Thesis:** *Machine Learning for Automatic Resource Management in the Datacenter and the Cloud UT Austin (Assistant Professor).*

40. Tianjun Zhang (PhD) [2018-2024]: **Thesis:** *System Architecture for Agentic Large Language Models Meta GenAI.*

41. Lianmin Zheng (PhD) [2019-2024]: **Thesis:** *Scalable and Efficient Systems for Large Deep Learning Models UCLA (Assistant Professor); xAI.*

42. Alan Zhu [2024-Present]: *Studying LLM personalization and agent adaptability in complex multi-agent environments.*

Masters Students

1. Raghav Anand [2019-2020]
2. Alex Cao [2026]
3. Anusha Dandamudi [2021]
4. James DeLoye [2025]
5. Aman Dhar [2019-2020]
6. Mohamed Elgharbawy [2023]
7. Edward Fang [2019-2020]
8. Tianren Gao [2021]
9. Daniel Ho [2019-2020]
10. Roy Huang [2025]
11. Samvit Jain [2017-2019]
12. Sona Jeswani [2019-2020]
13. Alvin Kao [2019-2020]
14. Shreyas Krishnaswamy [2023]
15. Alexander Kristoffersen [2023]
16. Alexander Ku [2019-2020]
17. Raguvir Kunani [2022]
18. Scott Lee [2019-2020]
19. Eric Leong [2022]
20. Debbie Liang [2022]
21. Eric Liu [2019-2020]
22. Yika Yujia Luo [2018-2019]
23. Kevin Miao [2022]
24. Ashwinee Panda [2019-2020]
25. Alan Pham [2022]
26. Aditya Ramkumar [2022]
27. Vedaad Shakib [2022]
28. Jiarui Shan [2023]
29. Chirag Sharma [2023]
30. Linyue Song [2022]
31. Vikranth Srivatsa [2023]
32. Hari Subbaraj [2019-2020]
33. Victor Sun [2019-2020]
34. Nikita Vemuri [2019-2020]
35. Derek Wan [2021]
36. Rachel Xin [2025]
37. Flora Xue [2019-2020]
38. Edward Yam [2022]
39. Andy Yan [2019-2020]
40. Fanjia Yan [2025]
41. Bohan Zhai [2021]
42. Corey Zumar [2017-2018]

Undergraduate Research Assistants

1. Daiyaan Arfeen
2. Avinash Arjavalingam
3. Michael Chau
4. Saurav Chhatrapati
5. Harrisson Constantinio
6. Anusha Dandamudi
7. Ziyi Dou
8. Rehan Durrani
9. Tianren Gao
10. Noah Golmant
11. Jennifer Grannen
12. Yijie Huang
13. Weston Hughes
14. Yuxin Ji
15. Henry Jin
16. Sun Ah Lee
17. Cathy Li
18. Michael Luo
19. William Ma
20. Sean Meng
21. Mong Ng
22. Aniruddha Nrusimha
23. Aditya Ramkumar
24. Allen Shen
25. Younjin Song
26. Ishaan Srivasta
27. Priya Sundaresan
28. Andrew Tan
29. Julie Yu
30. Anthony Yu
31. Bohan Zhai
32. Ziyun Zhao
33. Kevin Zhu
34. Tyler Zhu
35. Yitian Zou

Teaching

- **Introduction to Machine Learning [CS189/289A]:** Joseph co-teaches this large upper-division and graduate machine learning course at UC Berkeley covering foundational and modern ML topics including probability, regression, neural networks, optimization, transformers, and generative AI. Taught Fall 2025 and Fall 2026 (co-taught with Narges Norouzi). For more information visit `https://eecs189.org`.
- **Principles and Techniques of Data Science [Data100]:** Joseph developed and teaches the large upper division data science class at UC Berkeley. This class covers topics ranging from data cleaning and visualization to machine learning and hypothesis testing with a focus on real-world data and problems. Students from across campus take this class which routinely has more than 1000 students a semester. Taught Spring 2017, Fall 2017, Spring 2018, Spring 2020, Spring 2021, Spring 2024, and Fall 2024. For more information visit `http://ds100.org`.
- **Conversations with Thought Leaders in Technology (CS198-100):** Joseph developed and co-hosted this class with Vik Singh external industry leader. This interactive zoom seminar class created in the middle of the COVID-19 pandemic leveraged the virtual format to invite world class technology leaders to speak with students about their careers and the lessons they learned. In this class, we heavily leveraged slack and zoom to have student write and vote on questions that were then used to drive the interactive interview. Speakers in this series included Henry Schuck, Sanjay Poonen, Mamoon Hamid, Archana Agrawal, Hilary Mason, and Nick Caldwell.
- **AI Systems Graduate Seminar [CS294-162]:** Joseph developed and teaches this graduate seminar covering key topics in machine learning and AI systems research, combining paper reading with structured discussions and class projects. Taught Spring 2019 (as CS294-159), Fall 2019, Spring 2022, and Fall 2023 (LLM Edition). For more information visit `https://ucbrise.github.io/cs294-ai-sys-sp22/`.
- **Foundations of Data Science [Data 8]:** Joseph co-taught this large introductory data science course at UC Berkeley. Taught Spring 2023.
- **Introduction to Databases [CS186]:** Joseph co-taught this 500+ student class with Joseph Hellerstein and helped to update content and create a series of new lectures covering concepts at the intersection of databases and data science. Taught Spring 2016.

Workshops

- **[2019] ACM-IMS Interdisciplinary Summit on the Foundations of Data Science:** I co-organized this event and moderated the panel: Deep Learning, Reinforcement Learning, and Role of Methods in Data Science.
- **[2019] SOSP ML Systems Workshop** I co-organized the SOSP learning systems workshop. (`http://learningsys.org/sosp19/index.html`)
- **[2019] NeurIPS ML Systems Workshop** I co-organized the NeurIPS learning systems workshop. (`http://learningsys.org/neurips19/`)
- **[2019] Workshop on Human In the Loop Learning (HILL)** I co-organized the first workshop on Human in the Loop Learning co-located with ICML
- **[2018] NIPS ML Systems Workshop:** I co-organized the NIPS learning systems workshop. (`http://learningsys.org/nips18/`)
- **[2018] NextProf Nexus Workshop** I co-organized this three day workshop intended to strengthen and diversify the next generation of academic leaders in engineering. (`http://nextprof2018.engin.umich.edu/nexus/`)

- **[2017] SOSP ML Systems Workshop** I launched and co-organized the first SOSP learning systems workshop. (`http://learningsys.org/sosp17/`)
- **[2017] NIPS ML Systems Workshop** I co-organized the NIPS learning systems workshop. (`http://learningsys.org/nips17/`)
- **[2016] ICML ML Systems Workshop** I co-organized the ICML learning systems workshop. (`https://sites.google.com/site/mlsys2016/`)
- **[2015] NIPS Learning Systems Workshop** I help co-organize the NIPS learning systems workshop. (`http://learningsys.org`)
- **[2014] DIMACS Workshop Organizer** I organized the DIMACS workshop on the Systems and Analytics of Big Data (`http://dimacs.rutgers.edu/Workshops/Analytics/`)
- **[2013] NIPS Workshop Organizer** I helped organize the third annual NIPS "Big Learning: Algorithms, Systems, and Tools" workshop. (`http://biglearn.org`)
- **[2012] NIPS Workshop Organizer** I helped organize the second annual NIPS "Big Learning: Algorithms, Systems, and Tools" workshop. (`http://biglearn.org`)
- **[2011] NIPS Workshop Organizer** I organized and led the workshop entitled "Big Learning: Algorithms, Systems, and Tools for Learning at Scale" For more information visit the workshop website `http://biglearn.org`
- **[2009] NIPS Workshop Organizer** I organized and led the first NIPS BigLearn workshop entitled "Large-Scale Machine Learning: Parallelism and Massive Datasets." For more information visit the workshop website `http://www.select.cs.cmu.edu/meetings/biglearn09`

Funding Activities

- **[2019]** NSF National Science Foundation, CAREER: Towards the Design of Models and Systems for Efficient Prediction. UCB Award ID: 047303-001. PI: Joseph Gonzalez (UC Berkeley). Total: $\$450,648$.
- **[2018]** NSF National Science Foundation, ECDI: Secure Fog Robotics Using the Global Data Plane. UCB Award ID: 045561-001. Co-PI: Joseph Gonzalez (UC Berkeley). Total: $\$2,000,000$.
- **[2018]** NSF National Science Foundation, Expedition Award: Secure, Real-Time Decisions on Live Data. UCB Award ID: 044393-001. Co-PI: Joseph Gonzalez (UC Berkeley). Total: $\$9,999,999$. Project length: 5 years.
- **[2017]** DOD Advanced Research Projects Agency, Hierarchical Learning-Guided Automatic Model Discovery. UCB Award ID: 042431-001. Co-PI: Joseph Gonzalez (UC Berkeley). Total: $\$3,811,944$. Project length: 4 years.
- **[2017-present]** Raised gift funding for RISELab from Alibaba, Amazon, Ant Financial ARM, Ant, Capital One, Ericsson, Facebook, Futurewei, GE, Google, Hauwei, IBM, Intel, Microsoft, Nvidia, Scotiabank, Splunk, and VMware. Roughly $\$5,000,000$ annually.
- **[2013]** Raised 6.75M in series A funding for the GraphLab Inc. startup.
- **[2010]** Applied for and was awarded a grant to have early access to the Intel Single-chip Cloud Computer (SCC) as part of the Many-core Applications Research Community.
- **[2008 - 2009] Helped Lead a DARPA Interdisciplinary Sciences and Technology Study (ISAT) Group** to investigate the future of parallel machine learning from an interdisciplinary perspective I also participated in the final Woodshole annual ISAT meeting to prepare a proposal for the DARPA director.
- **[2008]** Applied for and was awarded funding for (BAA 08-34) "Machine Learning and AI in the context of Multicore and Cluster Computing."

University Service

- **[2025-2026] CS Division Vice Chair for Graduate Matters, UC Berkeley EECS:** Responsible for overseeing graduate student affairs in the Computer Science division, advising on graduate student petitions, academic standing, and policy matters.
- **[2025-2026] EECS Graduate Matters Committee:** Co-chair of the committee overseeing graduate curriculum, pedagogy, and student affairs for the EECS department.
- **[2025-2026] DIVCO (Divisional Council) Member — Elected (UC Berkeley Academic Senate, Berkeley Division):** Elected by the faculty to serve on the Divisional Council, the executive committee of the Berkeley Division of the Academic Senate, participating in campus-wide academic policy. Ran for this seat to help the campus address the looming challenges and opportunities of AI in higher education.
- **[2023-2024] EECS Undergraduate Study Committee (CS co-Vice Chair):** Co-chair of the EECS Undergraduate Study Committee, responsible for overseeing undergraduate curriculum and academic policy matters for the Computer Science division.
- **[2024-2025] EECS Undergraduate Study Committee (Member):** Continued service on the EECS Undergraduate Study Committee.

Reviewing

- **[2027] MLSys General Chair**
- **[2026] NeurIPS Area Chair**
- **[2026] ICLR Area Chair**
- **[2026] ICML Program Committee**
- **[2026] ECCV Reviewer**
- **[2026] CVPR Reviewer**
- **[2026] ACL Rolling Review (ARR) Reviewer**
- **[2025] ICLR Reviewer**
- **[2024] NeurIPS Datasets and Benchmarks Track Area Chair**
- **[2024] ACL Rolling Review (ARR) Reviewer**
- **[2024] COLM Reviewer**
- **[2023] ICLR 2024 Area Chair**
- **[2023] NeurIPS 2023 Reviewer**
- **[2022] ICLR 2023 Area Chair**
- **[2021] OSDI Program Committee**
- **[2021] SysML Program Committee**
- **[2020] SysML Conference Workshop Committee**
- **[2020] EuroSys Program Committee**
- **[2019] SOSP Program Committee**
- **[2019] MLSys (formerly SysML) Program Committee**
- **[2018] ICML Area Chair**
- **[2018] LXAI Workshop Program Committee**
- **[2018] MLSys (formerly SysML) Program Committee**
- **[2018] Sigmod Program Committee**
- **[2017] AIStats Senior Program Committee**
- **[2016] VLDB PC Member (Best Reviewer Award)**
- **[2017] EuroSys PC Member**
- **[2016] ICML Reviewer**
- **[2016] AISTATS Reviewer**
- **[2014] OSDI External Review Committee**
- **[2014] HotCloud Review Committee**
- **[2014] ICML Reviewer**
- **[2013] Transactions on Pattern Analysis and Machine Intelligence**
- **[2013] NIPS Reviewer**
- **[2013] Super Computing Reviewer**
- **[2012] Parallel Computing Reviewer**
- **[2010] ICML Reviewer**
- **[2009] JMLR Reviewer**
- **[2007] JMLR Reviewer**
- **[2007] IPSN Reviewer**

Commercial Involvement

- **Advisor to Inferact (2026-Present):** I serve as an advisor to Inferact, a startup commercializing vLLM that launched in January 2026 with $150M in seed funding.
- **Advisor to Letta Inc. (2024-Present):** I serve as an advisor to Letta, a UC Berkeley Sky Lab spinout founded by Charles Packer and Sarah Wooders to build stateful LLM agent infrastructure (based on the MemGPT research project).
- **Advisor to Genmo Inc. (2023-Present):** I serve as an advisor to Genmo, a generative video AI company co-founded by my former student Paras Jain.
- **Co-founder of RunLLM (formerly Aqueduct, 2020-Present):** I co-founded this company with former students and I am head of product.
- **Advisory Board for Anyscale Inc. (2019-Present):** I am an advisor for Anyscale.
- **Advisory Board for DeepScale Inc. (2017-2019):** I was an advisor for DeepScale and was involved in technical discussions around vision and systems for autonomous driving.
- **Turi Inc. (2013-2016):** I co-founded Turi Inc. (formerly GraphLab Inc.) to commercialize my research and help lead the initial technology roadmap. When Turi was acquired by Apple Inc. in July of 2016 it had grown to over 50 employees and had raised nearly 25M in venture capital.
- **Yahoo! Research (2011):** Developed the next generation of the GraphLab abstraction to enable large-scale machine learning on natural graphs derived from social media and web-content. *(Alex Smola: smola@yahoo-inc.com)*
- **AT&T Labs Research (2007):** Developed models for statistically assessing DSL quality from limited noisy data. *(Steven Phillips: phillips@research.att.com)*
- **Intern at ADAPT (2006)** Worked on an automated AdWords auction agent. I developed and implemented models for assessing word value. *(Alex Bäcker: alex@caltech.edu)*
- **Microsoft Developer Internship (2005):** Worked with MSN Search team developing techniques to use behavioral information to identity search spam. *(Greg Hullender: greghull@windows.microsoft.com)*
- **Caltech Research Fellowship (2004):** Developed a new query-less search technology that uses prior reading interests to identify novel documents. *(Alex Bäcker: alex@caltech.edu)*
- **NASA Jet Propulsion Labs Fellowship (2003):** Developed a new algorithm for efficiently evaluating line-of-sight on digital elevation maps at JPL. *(Robert Chamberlain: rgc@jpl.nasa.gov)*

Publicly Released Software

- **vLLM:** High-throughput and memory-efficient LLM inference and serving engine; one of the most widely used open-source LLM serving systems. `https://github.com/vllm-project/vllm`
- **SGLang:** A structured generation language and high-performance serving system for LLMs and vision-language models. `https://github.com/sgl-project/sglang`
- **Arena (formerly LMSYS Chatbot Arena):** An open platform for evaluating LLMs through crowdsourced human preference judgments; the leading open benchmark for LLM quality. Originally launched as Chatbot Arena under the LMSYS research organization at UC Berkeley, spun off into Arena Intelligence Inc. in April 2025 ($600M valuation), raised a $150M Series A in January 2026 (valuation $1.7B), and rebranded to Arena. `https://arena.ai`
- **RouteLLM:** A framework for cost-effective LLM routing, dynamically selecting between strong and weak models based on query difficulty. `https://github.com/lm-sys/RouteLLM`
- **Gorilla / Berkeley Function Calling Leaderboard (BFCL):** Gorilla is an LLM trained to generate accurate API calls; the associated BFCL is the leading benchmark for evaluating LLM tool-use and function-calling ability, now in V4 with agentic evaluation. `https://gorilla.cs.berkeley.edu`

- **MemGPT / Letta:** A framework for building stateful LLM agents with persistent, self-managed memory; originally published as MemGPT and rebranded to Letta, which was spun off into Letta Inc. (2024). `https://github.com/letta-ai/letta`
- **LEANN:** A low-storage vector index enabling RAG on personal devices with 97% storage savings over standard indexes while maintaining high recall; enables private, on-device semantic search over millions of documents. (MLSys 2026) `https://github.com/yichuan-w/LEANN`
- **Alpa:** A compiler for distributed deep learning. `https://github.com/alpa-projects/alpa`
- **SkyPlane:** Fast bulk data transfers across regions and clouds. `https://skyplane.org`
- **RLlib:** A distributed reinforcement learning system. `https://docs.ray.io/en/master/rllib.html`
- **Tune:** A system for distributed hyperparameter search and model development. `https://docs.ray.io/en/latest/tune.html`
- **Modin:** A system for scalable execution of Pandas dataframes. `https://modin.readthedocs.io/`
- **Clipper Prediction Serving System** A system for serving predictions across a wide range of machine learning frameworks. `https://clipper.ai`
- **Opaque:** Secure analytics on top of Apache Spark. `https://github.com/mc2-project/opaque`
- **GraphX:** A graph computation framework built into the widely adopted Apache Spark open-source project. `https://spark.apache.org/graphx/`
- **GraphLab/PowerGraph** A sophisticated API for building parallel and distributed machine learning algorithms on top of multicore and cloud architectures. GraphLab generalizes the MapReduce abstraction to support iterative asynchronous computation on graph structured dependent data. `http://graphlab.org`
- **Distributed SplashBP** This library implements the SplashBP algorithm for factor graph inference in the distributed setting using MPI. `http://www.select.cs.cmu.edu/code/mpi_splash.tar.gz`